International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory

Lab 6 – Operator Overloading (Group 1)

Date: 09.10.2026 **Semester:** B.Tech 3rd Semester **Group:** 1

Instructions

- Write and execute the C++ program for each problem.
- Use classes and objects to implement the required operations.
- Overload only the operators specified in each question.
- Display clear and meaningful output.
- Do not modify original objects unless explicitly required.
- Save all programs in your assigned GitHub repository.
- The repository submission must be completed by 12:00 AM on the same day to receive marks.
- Submit the lab record in the prescribed format.

1. **Fraction Calculator – Addition and Subtraction**

   Create a class `Fraction` containing a numerator and a denominator. Overload the binary `+` and `-` operators to add and subtract two fractions.

   Simplify each resulting fraction using the greatest common divisor (GCD). Ensure that the denominator of the final result is positive.

   Hint: *Use a separate helper function to simplify the fraction.*

2. **Time Duration Calculator**

   Create a class `Duration` containing hours and minutes. Overload the binary `+` operator to add two duration objects.

   Normalize the result so that minutes are always less than 60. Return a new object and display both the original durations and the resulting duration.

3. **Book Price Ranking**

   Create a class `Book` containing a book title and price. Overload the `<` operator to compare two books by price.

   If the prices are equal, the book with the lexicographically smaller title should be considered smaller. Return a Boolean result.

4. **Unary Operator – Account Adjustment**

   Create a class `AccountBalance` containing a balance. Overload the unary `-` operator to return a new object whose balance is the negative of the original balance.

   Display both objects and verify that the original balance remains unchanged.

5. **Score Tracker – Prefix and Postfix Increment**

   Create a class `Score` containing an integer score. Overload both prefix and postfix `++` operators.

   Demonstrate the difference between `++obj` and `obj++` by storing their results in separate objects and displaying the values.

   Hint: *The postfix operator function uses a dummy `int` parameter to distinguish it from the prefix version.*

6. **Date Equality and Inequality**

   Create a class `Date` containing day, month, and year. Overload the `==` and `!=` operators.

   Two date objects are equal only if their day, month, and year are all equal. Display the results of both comparisons.

7. **Inventory Combination**

   Create a class `InventoryItem` containing product ID, unit price, and quantity.

   Overload the binary `+` operator to combine two objects only if their product IDs and unit prices match. The resulting object should contain the combined quantity.

   If the objects are incompatible, report the situation clearly. Do not modify either original object.

8. **Temperature Comparison**

   Create a class `Temperature` containing a Celsius value. Overload the `>` and `<` operators to compare two temperature objects.

   Additionally, overload the unary `-` operator to return a new object with the negated temperature value.

   Demonstrate all three operations.

9. **Matrix Addition**

   Create a class `Matrix` representing a 2×2 matrix. Overload the binary `+` operator to add two matrix objects element-wise and return a new matrix.

   Display both original matrices and the resulting matrix. Ensure that the original objects remain unchanged.

10. **Shopping Bill Operations**

    Create a class `Bill` containing the number of items and the total bill amount.

    Overload the binary `+` operator to combine two bills by adding their item counts and total amounts. Also overload the `>` operator to compare two bills by their total amount.

    Display the combined bill and the result of comparing the two original bills.